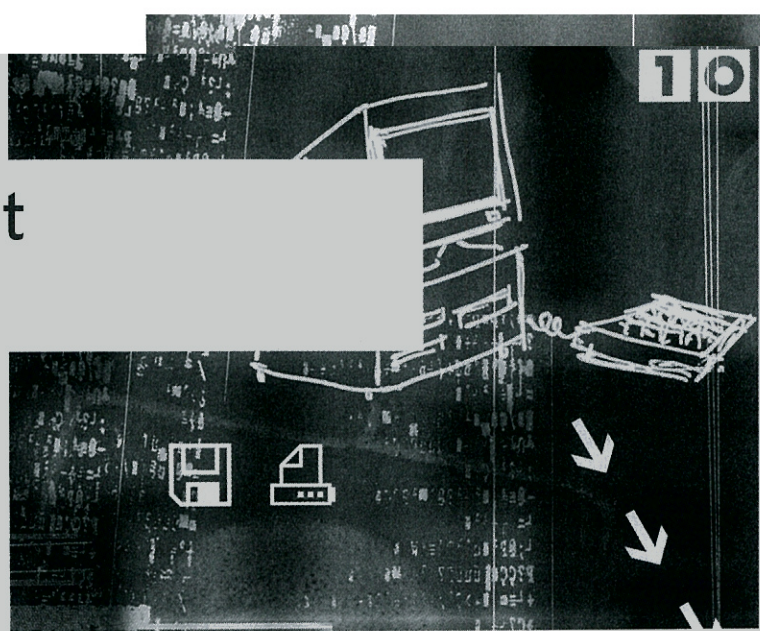


Envisionment



Aims

Envisionment is concerned with making ideas visible; with externalizing thoughts. Externalization can take all manner of forms: stories and scenarios, presentations, sketches, formal models, software prototypes, cardboard models and so on. Different forms of representation will be more or less useful at different stages in the design process and more or less effective for doing different things. A formal presentation of a design concept for a potential client will look quite different from a sketch of a screen layout intended to explore what something will look like. Envisionment is needed to represent design work to ourselves and to others. It occurs throughout development as the designer generates multiple design solutions and whittles them down to a final product. In this chapter we consider the principal envisionment techniques which (broadly speaking) use text and graphics; in the next chapter, we look at various forms of prototype and the presentation of ideas to clients. But first of all we review ways of thinking about the ideas to be externalized. After studying this chapter you should be able to:

- Exploit the possibilities of different media and metaphors in design concepts
- Use a variety of techniques for envisioning design problems and possible solutions
- Understand the role of concrete scenarios in envisioning design.

10.1 Exploring design concepts

Bill Verplank (Verplank, 2003) is an interaction designer who has been sketching and designing for many years. He argues that interaction design is 'design for human use' and focuses on three main things which he characterizes as

- How do you do?
- How do you feel?
- How do you know?

How do you do?

'How do you do?' is concerned with the ways in which we affect the world. Do you poke it, manipulate it, sit on it? For example, one distinction he highlights is between handles and buttons. Handles are better for continuous control (e.g. trombone), but buttons are better for discrete control (e.g. a piano keyboard). Handles leave you in control (e.g. opening a car door) whereas buttons are more likely to trigger something automatic (e.g. opening an elevator door).

How do you feel?

'How do you feel?' concerns how we make sense of the world and the sensory qualities that shape media. One distinction is Marshall McLuhan's 'hot' versus 'cool'. Marshall McLuhan wrote *Understanding Media* in 1964 and is famed for coining the phrases 'Global Village', 'age of information' and 'the medium is the message'. The book, which was reprinted in 1994, is a whirlwind tour through the media of his time and has much insight into how media would develop in our time. He introduced a distinction between 'hot media' which are more authoritative and exact and 'cool media' which are fuzzy and incomplete. Cool media invite more participation; they require the audience to fill in the gaps, to interpret. Hot media extend a single sense in high definition; they are filled with data. Photography is a hot medium because it is high fidelity, whereas a cartoon is a cool medium, low definition where we fill in the gaps. McLuhan's book is a fascinating if difficult read and he extends the ideas of hot and cool to all manner of concepts such as axes (hot), the 'city slicker' (hot), rural life (cool). Focusing on 'how do you feel' takes us into the areas of satisfaction, affect, enjoyment, involvement and engagement. In his introduction to the 1994 MIT Press edition of *Understanding Media*, Lewis Lapham characterizes McLuhan's ideas as illustrated in Table 10-1. Do not worry too much about understanding these dichotomies; use them as ways of thinking about 'how do you feel'.



Chapter 17
discusses
affect and
emotion

How do you know?

'How do you know?' concerns the ways that people learn and plan; how designers want users to think. For example, Verplank suggests that one choice is between maps and paths. Paths are good for beginners as they provide step-by-step instructions on what to do. Maps are good for understanding alternatives. They take longer to learn but are more robust and are good for expert skill. Maps offer the chance to take short cuts. Very often, of course, a given system or product will have to accommodate both.

TABLE 10-1 Leitmotifs from McLuhan's *Understanding Media*

Print	Electronic media
Visual	Tactile
Mechanical	Organic
Sequence	Simultaneity
Composition	Improvisation
Eye	Ear
Active	Reactive
Expansion	Contraction
Complete	Incomplete
Soliloquy	Chorus
Classification	Pattern recognition
Centre	Margin
Continuous	Discontinuous
Syntax	Mosaic
Self-expression	Group therapy
Typographic man	Graphic man

Source: Lapham (1994) in McLuhan, M. *Understanding Media: The Extensions of Man, New Edition*. Published and reprinted by permission of The MIT Press

**Challenge 10-1**

What characteristics do you feel belong to the text message as a medium?
Be creative!

10.2 Using metaphor to explore design



Chapter 25 on
navigation of
information
space

Paths and maps may be thought of as metaphors for the design of interactions. Different metaphors will lead to different conceptions and designs. Think of the idea of navigating an interactive system, for example. Many people immediately think that navigation is trying to get somewhere specific, but this is only one view (often called 'wayfinding'). We also browse around and explore. If we think of navigation as in a city then we think of roads and signposts, subways that take us invisibly from one part of the space to another, taxis to transport us, or buses that we have to learn about.

Considering one metaphor can stimulate creative leaps into other ways of thinking. For example, during a collaborative design meeting to look at the HIC interface as a 'rolodex' (Figure 10-1), a manual card index device that allowed people to rapidly flip through standard-sized cards, a 'gun barrel' design emerged (illustrated

Box 10-1 Navigation metaphors

Metaphors which might be useful for thinking about navigation have different associations. A wilderness, for example, is frightening, confusing, enchanting. A desert is intimidating and beautiful but has no landmarks. Such metaphors might encourage forging a path, enjoying the scenery and getting out. Wilderness and desert can be included in an overall landscape metaphor where different types of terrain represent different types of information. These metaphors encourage exploration by the user; the system provides a high-level metaphor, but users provide the more detailed structure themselves.

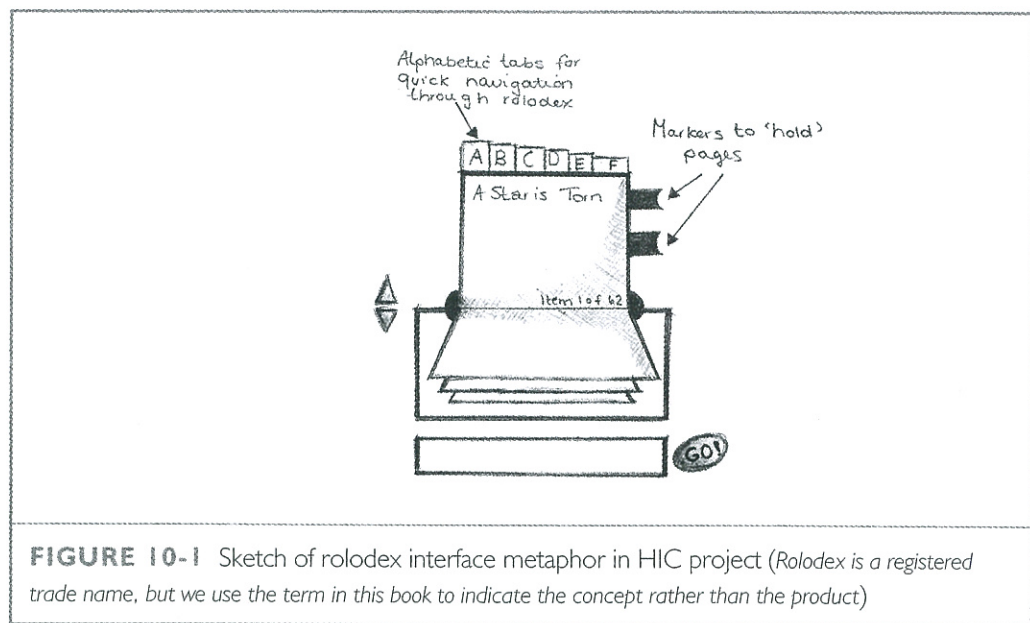
The night sky offers a different sort of space. To the human eye, it contains objects, clusters and patterns and it is very big. It supports the activities of mapping and identifying objects. However, there are relatively few types of object in the night sky (galaxies, stars, planets). It is the configurations and sub-types of these objects which are of interest. Science fiction concepts such as warp drives and worm holes can be used for quick and magic transportation to distant parts of the space. The open sea is another metaphor. It encourages a distinction between surface and depth. Thus it is natural to think of a lot of information as being hidden beneath the surface and only available for viewing if the user dives down. Currents can link continents and islands and take people to unexpected places. People can look for islands of information; archipelagos provide clusters. A museum has been structured to allow free roaming, yet is also structured to facilitate learning. A library is suitable for finding specific information. It is well organized and structured.

These metaphors are not meant to suggest that the interface to a particular product looks like a desert, wilderness or library (though sometimes such explicit interface metaphors can be useful); the idea here is to think about activities in different ways.

in the left-hand part of Figure 10-3). Although the gun barrel idea was rejected, the rolodex concept seemed to capture the idea of searching and retrieving results with an intuitive navigational style of flicking through the 'cards'. In this case the conceptual metaphor of the rolodex translated nicely into a visual representation. So here an interface metaphor seemed wholly appropriate and indeed was implemented as part of the first functional prototype.



Chapter 13,
Figure 13.3
shows the final
implementation



**Challenge 10.2**

Think of three different metaphors (or ways of thinking about) that could be used for a diary function on a personal digital assistant.

10.3 Finding suitable representations

Envisionment is fundamental to effective human-centred design, to enable designers to see things from other people's perspectives and to explore design concepts and ideas with others. Different representations of design ideas are useful at different stages for different people. They help with generation, communication and evaluation of ideas. A sketch 'on the back of an envelope' might be useful for generating an idea and expressing it to a colleague – but it is not so good for giving to a client.

There are many techniques that can be used to help develop an understanding of the design problem and to envision possible solutions. None of these techniques in themselves will lead to the perfect design, but they will all generate some kind of document or representation which can be used in the process of communicating with clients, users and colleagues. It is through communication that design solutions will arise, be evaluated, and (eventually) be transformed into a final product.

Which techniques are used on a particular project will depend on a number of factors: the working style of the development team, the type of project, the resources available and so on. In an ideal world, developers would use a wide variety of representations, but in a two-person company working on a project with a deadline of four weeks this may not be possible.

Choosing suitable representations for the task at hand is one of the skills of a designer; another is making good use of that representation. Representations work by suppressing unnecessary detail, thus ensuring that the significant features of some artefact or activity stand out. A good representation is accurate enough to reflect the features of the system being modelled, but simple enough to avoid confusion. It adopts a style of presentation which is suitable for its purpose. Scenarios are one sort of representation – and their use for this phase of design will be explored later in this chapter – but there are many others.

Consider the following example:

A car designer has been commissioned to produce a new luxury sports car. He or she doodles a few designs on paper and shows them to other designers on the team. They make some comments and criticisms and as a result changes are made. Finally the designer is satisfied with one of the designs and draws up detailed blueprints which are given to the firm's model maker. Scale models are produced and sent to Marketing and Sales for customer reaction. The scale models are also subjected to wind tunnel experiments to investigate the aerodynamics of the design and the results are used in a computer program which will calculate the car's speed and fuel efficiency.

The designer is using four different representations in at least four different ways:

- The original representations represent a clearing of the mind. In this case they are doodles and sketches which are used to *generate new ideas*, examine possibilities and prompt for questions.
- The blueprints given to the model maker and the scale model given to the Marketing and Sales Departments are suitable for *accurately expressing ideas* to others.
- The wind tunnel experiments show representations being used to *test ideas*.
- The computer model is used to *make predictions*.



Challenge 10-3

Which representations in the example above are being used to explore the problem? Which are being used to communicate ideas?

10.4 Sketches and snapshots

The art of sketching is something that all designers should practise. Ideas and thoughts can be quickly visualized – either to yourself, or to others – and explored. Many an idea has come to fruition ‘on the back of an envelope’. The millennium bridge across the River Thames in London was reputedly designed on a paper napkin in a restaurant. Designers do well to carry a sketch book with them so that inspiration can be quickly captured and preserved.



The full scenario is in Chapter 14, Section 14.2

Figure 10-2 shows two visualizations of the Edinburgh Festival scenario. A key design requirement is to display large amounts of data without the need for scrolling (if possible). In the sketches we can see that the designer has been exploring different ideas for displaying and searching through results of a search. The design principle underlying the designs is often referred to as ‘focus and context’. Ben Shneiderman has a mantra which he encourages designers to use: ‘overview first, zoom and filter, then details on demand’ (Shneiderman, 1998, p. 523). On the left is a ‘hyperbolic tree’ representation of the Festival scenario and on the right is a ‘cone tree’ representation. Notice how the hyperbolic tree is better for capturing the focus and context principles.

Individual snapshots of a design can be provided to show key moments in an interaction and are particularly useful for exploring the impact of a certain style or design. Snapshots can be single sketches, or frames, from a storyboard (see below) or they can be produced using software. In Figure 10-3 we can see two more examples from the HIC case study. Once again, the aim is to highlight the main result but also to provide a context of related search results. It is clear how the first snapshot has influenced the second, a ‘gun barrel’ concept.

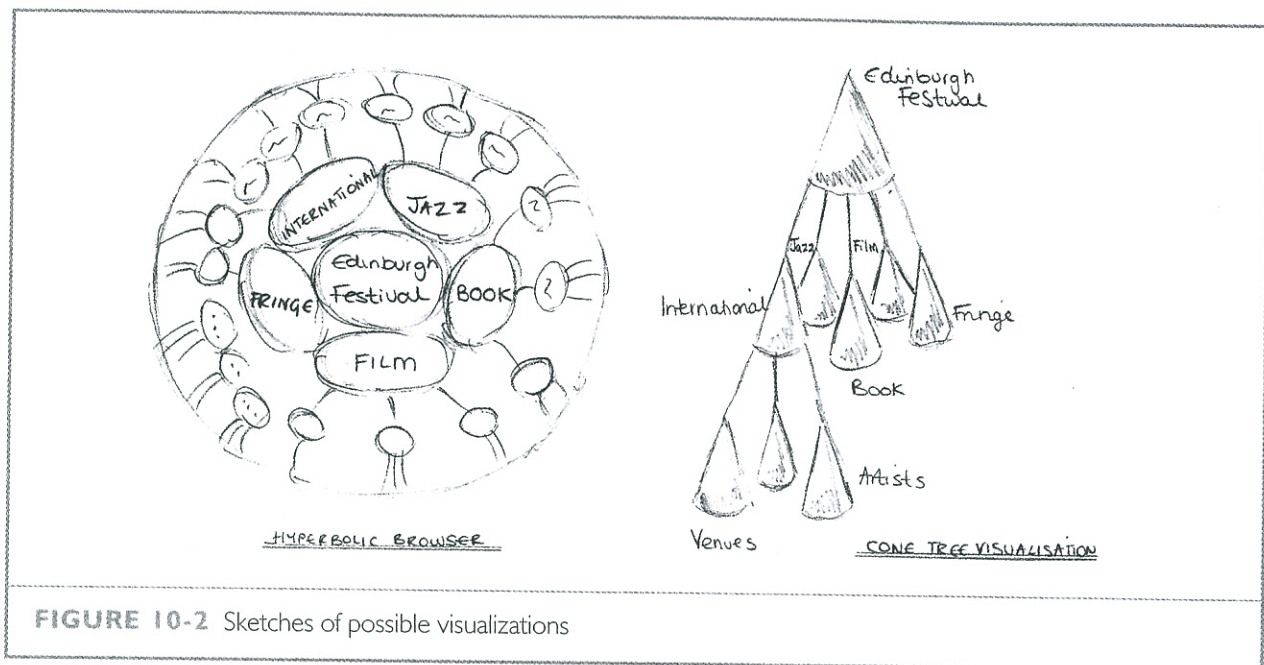


FIGURE 10-2 Sketches of possible visualizations

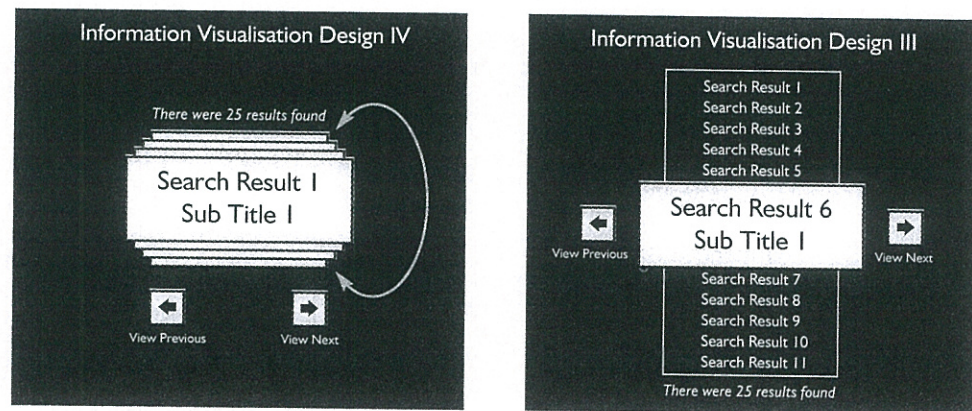


FIGURE 10-3 Snapshots of alternative designs for searching the HIC



Challenge 10-4

Sketch two different ways in which you might present information about tourist sites on a town's website.

10.5 Storyboards

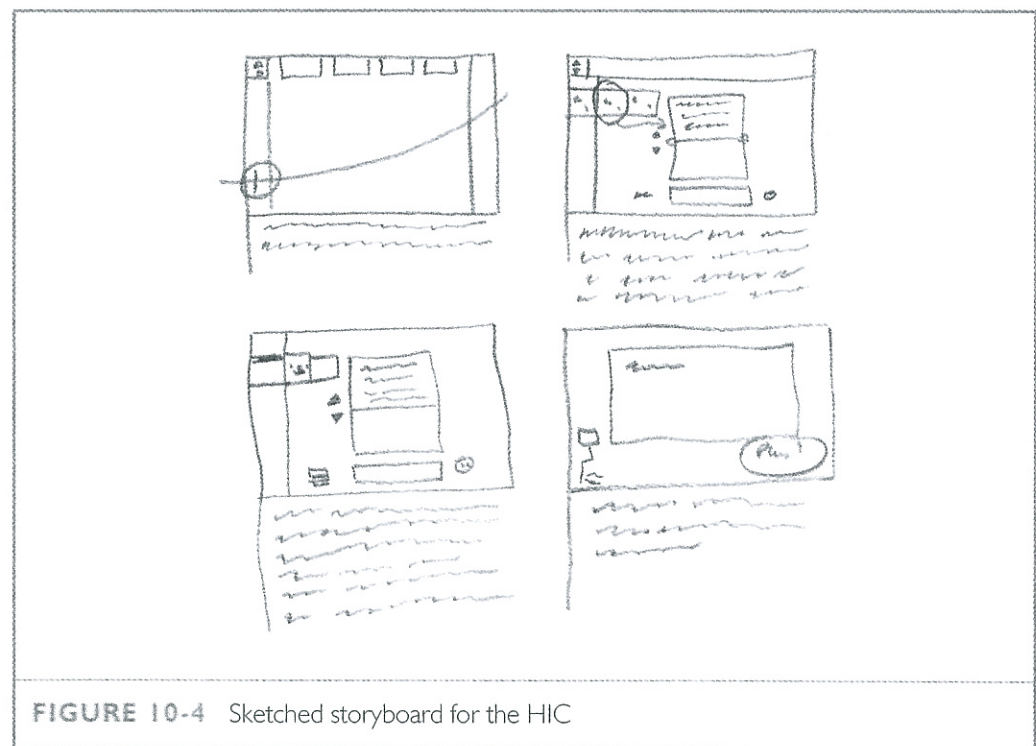
Storyboarding is a technique taken from film making – using a simple cartoon-like structure, key moments from the interactive experience are represented. The advantage of storyboarding is that it allows you to get a feel for the ‘flow’ of the experience. It is also a very economical way of representing the design – a single

page can hold 6–8 ‘scenes’. It is often helpful to sketch out a storyboard based around a concrete scenario. The two together are very helpful in working through design ideas with users.

Three main types of storyboarding are commonly found in interactive media design:

- **Traditional storyboarding.** A storyboard for a film would usually have some notes attached to each scene expanding on what will happen – this helps overcome the limitations of representing a dynamic experience in a static medium. For interactive systems, notes below each sketch usually contain the relevant steps from a scenario, and the sketches themselves are annotated to indicate interactive behaviour. This is the most usual form of storyboard if there is not a strongly multimedia flavour to the application.
- **Scored storyboards.** If the application has a lot of motion graphics the storyboard can be annotated – a sketch is annotated with appropriate notation and notes about, for example, type, colours, images, sound and other issues are attached underneath.
- **Text-only storyboards.** These are useful if the application has a lot of very complex sequences. You can specify what images appear, what text accompanies them, any accompanying media, general notes about tone, flow, etc.

Figure 10-4 shows a sketched storyboard for an application of the HIC to playing MP3 music files. The storyboard shows how an interaction might take place. It



exploits some ideas of marking movement using arrows and lines to indicate change. Figure 10-5 shows part of another storyboard, this time for a website designed to showcase a photographer's portfolio.

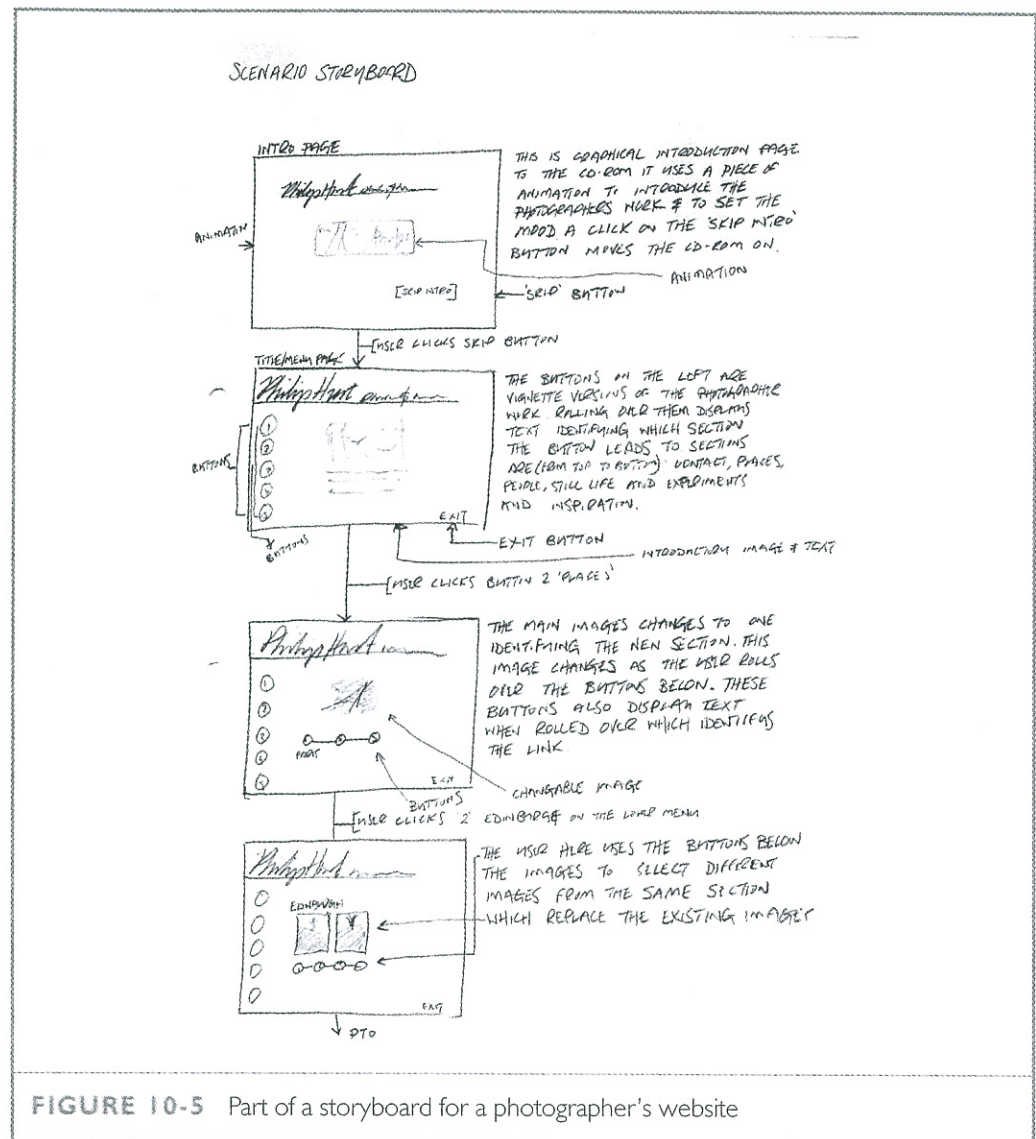
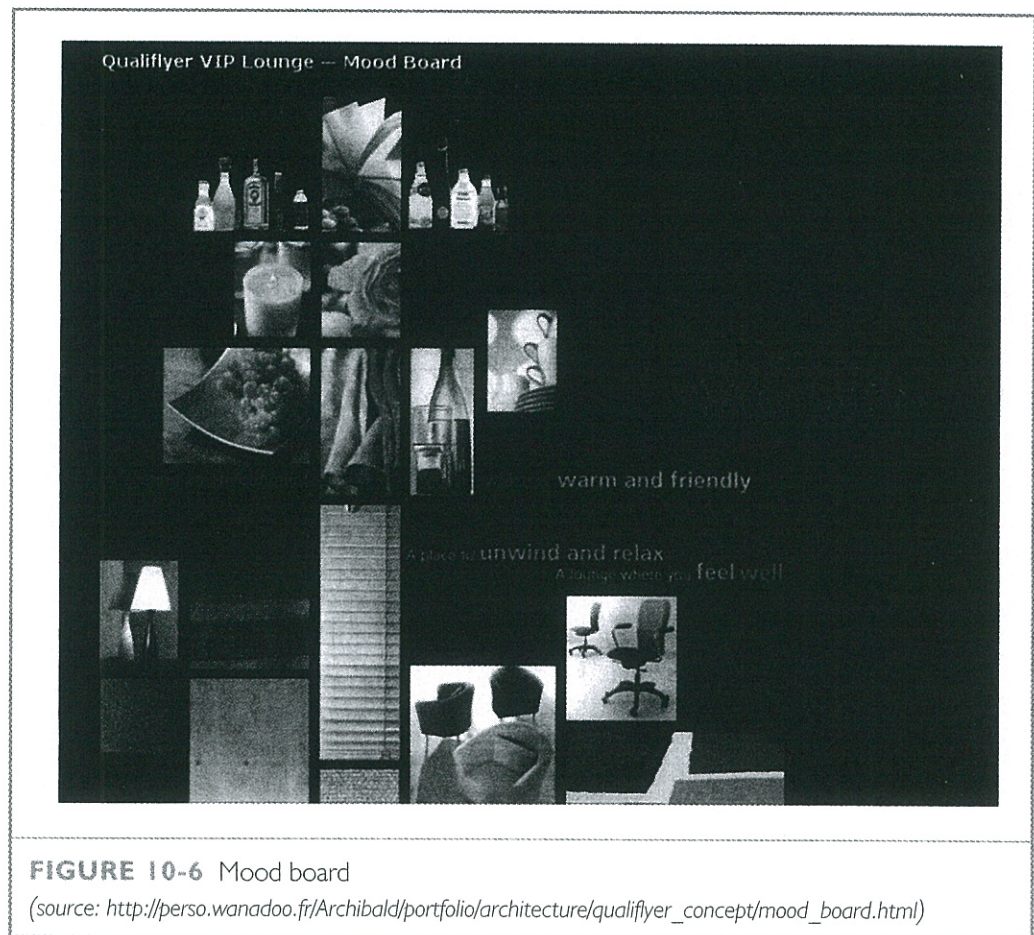


FIGURE 10-5 Part of a storyboard for a photographer's website

10.6 Mood boards

Mood boards are widely used in advertising and interior design. Quite simply you gather visual stimuli that capture something of how you feel about the design – photographs and other images, colours, textures, shapes, headlines from newspapers or magazines, quotations from people, pieces of fabric and so on. Attach the stimuli to a pinboard. Even thinking about their arrangement can

stimulate ideas. You can put pages from websites you like on mood boards. If you use Blu-tack or something similar then you can add and delete items as your thinking changes. Figure 10-6 shows an example.



The rule with mood boards is that ‘anything goes’. The point of the board is not to formally represent some aspect of the design, simply to act as inspiration – perhaps promoting a particular line of thought, or providing inspiration for a colour scheme. One technique is to get the client to create a mood board. This can give you an insight into the kinds of aesthetics that are likely to appeal to them. As a variation on the mood board concept, writing down adjectives that describe some aspect of the system can be useful.

10.7 Navigation maps

Navigation is a key feature for many systems. Navigation maps focus on how the user moves through the site or application. The aim is to focus on how users will experience the site. Each page in the site, or location in the application, is repre-



Chapter 25,
Section 25.3
discusses
navigation for
websites

sented with a box or heading and every page that can be accessed from that page should flow from it. A useful tip is to put in all flows possible (i.e. back and forwards from a page) as this will highlight sections where the user can get stranded. Navigation maps can usefully be redrawn many times through the project lifecycle, as poor navigational structure is one of the main reasons people turn off a website. The maps can be used with scenarios to 'walk through' particular activities and are a very good way of spotting poor aspects of design such as 'orphan pages' (pages which are not accessible) or dead ends.

Navigation is important in all manner of applications and products, not just websites. Figure 10-7 shows the navigation map for a mobile phone. More formal and more fully annotated maps can also be developed as illustrated in Figure 10-8. Arrows can be added to lines if the direction of a link is important. An excellent 'visual language' for describing websites is available at Jessie James Garrett's site www.jjg.net/ia/visvocab/.

Menu map

Items in *italics* depend on your subscription and your network operator.

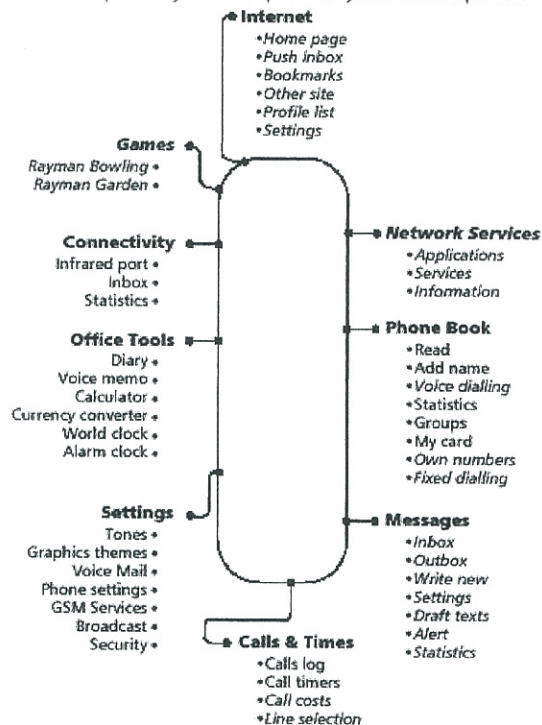


FIGURE 10-7 Mobile phone navigation map (source: *Trium phone manual*)



Challenge 10-5

Construct a navigation map for a website with which you are familiar - perhaps that of your university/college or employer. (If the site is very large, draw a partial map.) Are there any 'dead ends' or complicated routes to important information?

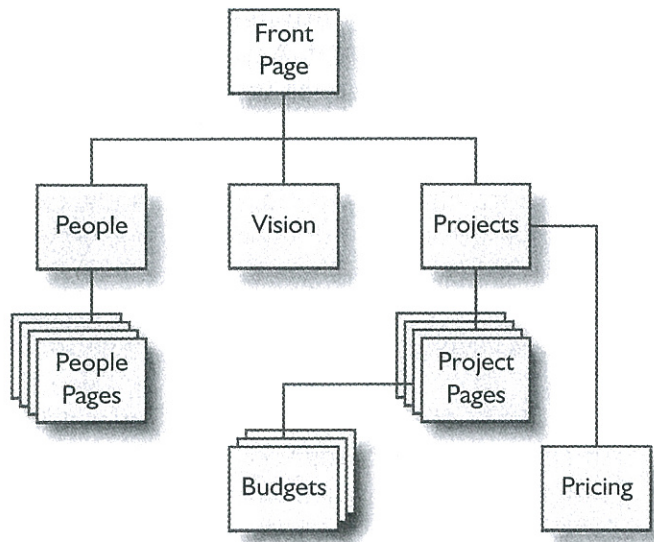


FIGURE 10-8 Navigation map for a website

10.8 Using scenarios in envisionment

At the end of Chapter 9 we saw how conceptual scenarios capturing the essence of new technologies can be used to embody requirements for a new design. In a scenario-based approach, the next stage is to enrich these scenarios with more concrete, contextual details that will prompt discussions and highlight the design decisions to be made. The scenarios should be discussed and evaluated at design team sessions and with the intended users. There are *degrees* of concreteness in scenarios. The most concrete forms are used to envision or evaluate specific interactions. While they are superficially easy to construct, there are a number of ways in which scenarios can be made more effective:

- Complement the scenarios with some of the more visual envisioning techniques.
- In a large design team, include real data and materials so people not directly involved can appreciate concrete details.
- Think hard about underlying assumptions.
- Include good characterization and develop a number of personas. If this is done well, members of the team start talking about the characters – ‘If you design it like that, what will happen when the grandmother tries to use it?’.
- Provide a rich contextual background – this grounds design decisions in real life, forcing the designer to think about practicality and acceptability.
- Team members can write their own concrete version of a conceptual scenario that reflects their particular concerns. These can be brought together and overlaps removed.



Personas were introduced in Chapter 2

The aim is to come up with a collection of scenarios that covers all the major uses and functionality of the product. It would be impossible to write scenarios for all possible variations in use and users, but those produced should cover:

- Interactions which are typical of a number of similar use situations
- Design issues which are particularly important for the focus of the project
- Areas where requirements are unclear
- Any aspects which are safety-critical.

Example 1: a simple scenario

Here is a scenario developed for the DISCOVER training application, illustrating some of the software features for tutors. Scenarios such as this were discussed at meetings between user representatives, designers and developers and helped to establish details of technical feasibility and fit with working practice at the training centres. Discussion of a similar scenario revealed substantial differences between expectations of the users that virtual fires (for example) would respond to trainees' behaviour in the environment and technological constraints which meant that only a small number of simple pre-programmed variations would be available.

Terry is a tutor at an oil industry training centre. He is familiar with the DISCOVER VR training system, which he is using for a short practice session with two senior staff on an oil platform. They are shortly to be assessed for a safety-management certificate. The staff have used DISCOVER at the training centre but are now located back at the platform in the North Sea. The three men have already run up DISCOVER in its multi-user mode and have established audio communication, using headsets. Terry uses the drop-down menu to select the trainee Sten to be monitored. By default, his viewpoint is slightly above and behind the head of Sten's avatar, and as Sten moves his avatar through the simulated oil platform Terry's viewpoint changes accordingly. Terry can also move viewpoint independently by selecting the compass icon. Terry speaks to the trainees to give initial instructions for the role-playing scenario - there is a fire in level 2, and the team are to search for crew members who may be trapped or overcome by smoke. The trainees set off, and after 30 seconds or so Terry moves his viewpoint to look behind Sten and observes that Trond (or rather his avatar) is following directly. Terry decides to complicate the situation. He clicks on a spot on the plan of the rig displayed in a window at the bottom of his screen and then on the fire icon from the tutor's palette to set off another fire in the mess hall. A dialogue box appears asking him to specify the scale of the fire and the amount of smoke, presenting the options as radio buttons ...

Structuring scenarios in larger design projects



Chapter 8
discusses
scenarios

As already shown in Chapter 8, larger projects and collections of scenarios need a degree of structure to keep them organized. A set of structured scenarios were developed for the MP3 function of the HIC. Following brainstorming sessions, analyses of existing MP3 players and the collection of various stories of the MP3 in use, five scenarios were developed. 'How does that song go again?' is described in full in Section 8.3. The others were:

Title	Activities	Rationale
'Is it on the Web?'	Searching for MP3 audio content on the World Wide Web; downloading and playing back MP3 tracks; file management. Bookmarking favourite websites for future use.	The area of concern is access to Web resources via the HIC. The scenario describes connecting to a Web resource (a website) and downloading MP3 content to the HIC. This involves file management functions including saving and deleting.
'It's my turn now'	Inspecting e-mail; downloading attached files; modifying HIC preferences.	The scenario takes the HIC's MP3 functions as a focus for the kind of conflicts which may arise when several users want to access the HIC at once - to use either the same domain or different domains. Protagonists here access the HIC's functions via personalized user profiles. There is also a description of how access to particular domains of the HIC can be blocked for selected users.
'Play me something while I cook'	Transferring MP3 files to the HIC; saving them to permanent memory; organizing/classifying MP3 files; searching for files by different criteria.	At least two broad areas of functionality are depicted in the scenario: the business of loading new MP3 content into the HIC via different media (including removable discs and the Web); and organizing MP3 files in the HIC using facilities for classifying by different criteria. There is also an account of simple search activities via user input.

Title	Activities	Rationale
'Be our DJ this evening'	Making compilations of MP3 tracks (play lists) within the HIC, and storing them. Editing them. Playing them back in different track orders.	Here the HIC's MP3 functions are used like a jukebox, to provide music for a party. The scenario deals in fairly general terms with the activities involved in making lengthy compilations of MP3 tracks from content already within the HIC.

10.9 Exploring the design space

When writing, thinking about or using scenarios, issues will arise about design features. This is particularly so when reviewing scenarios and other design representations with potential users. It is vital to do this. Very often it is only when people see some sort of concrete representation of the new system and how it will fit (or not) with their lives that they are able to comment meaningfully. Issues may concern the current state of affairs or a future designed position. We have seen how these issues can be highlighted through adding end notes to the scenario descriptions, and notes can also be appended to storyboards and so on. In many cases it will be necessary to revisit the requirements, which in turn may entail further analysis of the current situation.

John Carroll (e.g. Carroll, 2000) points out that in design work there are often trade-offs to consider and a design feature may lead to both positive and negative outcomes. He recommends listing positive and negative features of a design alongside a feature, as 'claims'. We have found it useful to identify neutral design features too. One way of highlighting design issues is to undertake a walkthrough using the concrete scenarios to drive the thinking of the designer. Scenarios are very effective at forcing issues into the open, so that claims about designs can be articulated, documented and evaluated.

Refer back to Section 8.3 and consider paragraph P4 of the MP3 scenario and endnotes 8 and 9. How should search results be displayed and selected? Just one of the many decisions associated with this activity is the size of the font used to display artists' names and track titles. Unlike other domains for the HIC such as News, the MP3 domain demands that quite a lot of text is displayed so that the user can make a selection. So font size is a key design issue.

Some of the positive and negative features of using a large font size are:

- It can be seen from further away (positive).
- It takes up valuable screen space (negative).
- It means fewer tracks can be displayed (negative).



Carroll's formulation for claims can be found in the advanced evaluation material in Chapter 27

There are many variations of techniques that can be used to focus attention on design issues. Listing the actions that people will have to take in order to accomplish a goal using a specific design and identifying the positive and negative aspects of a prototype is one way. Another is to list the options against the design criteria. This technique, known as QOC (Questions, Options, Criteria), is illustrated in Figure 10-9. QOC was developed by Alan MacLean and his colleagues at Rank Xerox (MacLean *et al.*, 1991). The criteria against which the options are ranked need to be developed as part of the design brief and requirements gathering process.

10.10 An outline envisionment process

Here is a suggested series of steps for the envisionment phase, pulling together the wide-ranging material in this chapter.

1. Review requirements and conceptual scenarios
- 2a. Develop representations of your design ideas. At a minimum these should include concrete scenarios, storyboards developing the main interaction sequences, and snapshot sketches of key screens or other aspects of the product. More complex or creative projects will benefit from the other

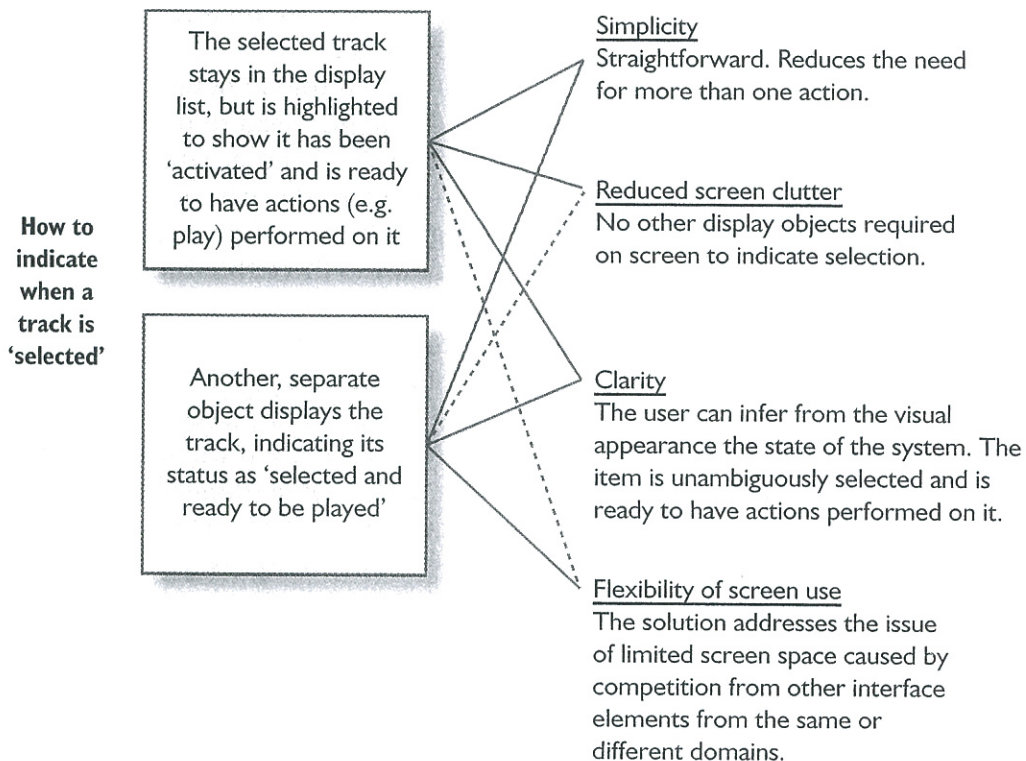
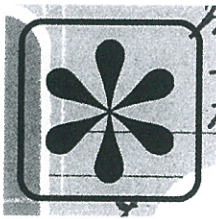


FIGURE 10-9 An example of the QOC (Questions, Options, Criteria) claims method

techniques discussed. Large developments in particular will benefit from the formal structure for concrete scenarios described above.

- 2b. If your product is a new one, experiment with different metaphors and design concepts through your representations.
- 2c. Your intended users should be involved in exploring design ideas throughout wherever possible.
3. Resources permitting, explore and document detailed design decisions using a method such as claims analysis or QOC.
4. Reconsider requirements in the light of the developing design, and carry out supplementary analysis where gaps in your background information have been uncovered.



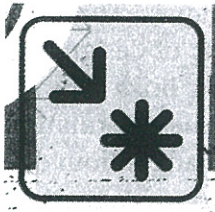
Summary and key points

These are just a few of the many possible envisionment techniques. There are books full of interesting and novel ways of representing aspects of design. For example, 'mind maps' list the main concepts of a design, showing links between them, flow diagrams show the movement of some items or some information through a design, and transition diagrams show how a system changes from one state to another. We discussed an example of a car designer who used sketches, and others in the design process who used scale models, blueprints and so on. We also argued that getting an appropriate representation – appropriate for a particular purpose in a particular context – is important. The different representations will be used alongside other techniques for helping to generate ideas such as brainstorming and other forms of collaborative design. For example, we have found sketching on a whiteboard very effective for a design meeting, as is using a flipchart where pages can be torn off and stuck to the wall. Writing ideas on Post-it notes and sticking them on the walls of a design room is another technique. They can be rearranged to show different collections and 'affinities'. Collaborative writing where a group all work on a single document using a computer and data projector to display the results on a screen can be very effective in producing draft documents.

A key feature of design and of the techniques described here is *not* to sit staring at a blank piece of paper. Getting inspiration from magazines, websites, software systems, other people, similar systems or products and so on, and externalizing ideas through envisionment techniques, is the first step in design. We have also seen in this chapter how thinking about the qualities of different media and underlying metaphors can play a part.

As we noted at the beginning of this chapter, one major envisionment technique remains – prototyping. This is discussed in the next chapter.

- Considering Verplank's design dimensions, the qualities of different media as suggested by McLuhan and underlying metaphors can help to stimulate design thinking.
- Envisionment – the making concrete of design ideas – is a key feature of design. All aspects of the system can and should be envisioned: concepts, functions, structure, interactions.
- Envisionment aids the generation, communication and evaluation of ideas.
- Users should take an active part in envisionment wherever possible – the process allows essential feedback from users themselves and clients.
- Techniques include storyboards, different forms of sketch, mood boards, navigation maps and concrete scenarios.
- Concrete scenarios in particular can be analysed using techniques such as claims analysis and QOC to represent design trade-offs so that decisions can be clearly seen.



Further reading

Browsing the design section of a good bookshop will find numerous books containing ideas to stimulate creativity – which you find helpful is very much an individual preference. Equally, the business section will offer a wide range of published material for enhancing the generation of ideas in group meetings.

McLuhan, M. (1994) *Understanding Media: The Extensions of Man*. MIT Press, Cambridge, MA.

This is not a book for the faint-hearted, nor for those who want some simple answers. It is something of a psychedelic tour through a 1960s view of the coming age of media. It makes you think.

Bill Verplank is at www.billverplank.com

MacLean, A., Young, R., Bellotti, V. and Moran, T. (1991) Questions, options and criteria: elements of design space analysis. *Human-Computer Interaction*, 6, 201–251.

The original paper describing the elements of QOC. Includes analyses of existing designs and supporting data about the concepts and arguments used by designers during design discussions.

Buckingham Shum, S., MacLean, A., Bellotti, V. and Hammond, N. (1997) Graphical argumentation and design cognition. *Human-Computer Interaction*, 12, 267–300.

Presents an extensive analysis of the use of QOC. The authors argue that the method provides most support when elaborating poorly understood design spaces, but is a distraction when evaluating well-defined issues.



Comments on challenges

Challenge 10-1

Personal, direct, fun, impromptu, young, quick, pervasive, hot ... occur to me, but there are almost infinite possibilities here. There are no right or wrong answers; it depends on your own perceptions and feelings.

Challenge 10-2

Thinking of physical diaries is one way to come up with possible metaphors: pocket diaries, desk diaries, the diary of personal events that you might keep, the diary of future events in an area such as sports, a diary of events for a local society or club. This might lead to the metaphor of a pocket diary in which the interface had 'pages' that could be 'turned'. The diary of events leads to a wallchart type of display with the days of the month laid out. The personal diary might lead to ideas such as secrecy and security, being able to write free-flowing text rather than having a regimented date- and event-driven design that a business diary suggests.

Challenge 10-3

The sketches and doodles are being used to explore the problem space. The computer simulation is also being used in this way and so the scale model put into the wind tunnel is an essential part of that representation. Both the blueprints and the scale model sent to Marketing are being used to communicate ideas. The blueprints and the scale model are both used for communication, but the blueprints are inappropriate for communicating with Marketing. Marketing people are interested in the physical shape of the design, but the model maker requires a more precise description of the designer's ideas in the form of blueprints. Also notice that the representations must be accurate enough for their purpose, highlighting the important features but ignoring the irrelevant aspects. In the wind tunnel, the interior design of the car is unimportant, so the scale model takes no account of this.

Challenge 10-4

Many different ideas are possible here. Examples could include a simplified interactive map, a text listing by category, a montage of clickable images, etc. Remember that what you are doing here is trying out ideas, so don't spend time adding too much detail or creating a work of art.

Challenge 10-5

No specific comments here, but make sure that you have shown the direction of links on the map where they are important. If there are multiple links to the same target, the map can be simplified by adding a note such as 'all pages link back to home page' and omitting the links themselves.



Exercises

1. Write a scenario for the home control device introduced in Challenge 10-9. You should use the formal structure introduced in this chapter. Choose three of the design features from the scenario and list the positive and negative aspects as described in Section 10.9.

2. Mood boards are usually constructed out of physical materials, but they can also be made in software. Using any software application that allows the inclusion of visual, audio and text components, make a mood board to explore the following concepts:

- A website for seaside holidays targeted at one-parent families
- A website for adventure holidays targeted at active, affluent over-sixties.

This can be done as a single-person exercise, but works best as a small group project.

3. (More advanced) We argue strongly in this book for users to be involved as closely as possible in the envisionment process. Develop some bullet points setting out a counter argument – that designers ‘know best’.